

Functions, Function Notation, and Slope

Directions: Show all work. No calculators.

1. If $f(x) = 3x - 7$, find $f(-4)$.

2. If $g(x) = x^2 - 4x + 6$, find $g(5)$.

3. If $h(x) = -2x^2 + 5x - 1$, find $h(-2)$.

4. If $p(x) = 8 - 3x$, find $p(6)$.

5. If $r(x) = 3x + 5$, solve $r(x) = 20$.

6. If $s(x) = 4x - 7$, solve $s(x) = 9$.

7. If $t(x) = 6x + 1$, solve $t(x) = 14$.

8. If $u(x) = -5x + 8$, solve $u(x) = -17$.

9. Find the slope of the line through $(-4, 2)$ and $(2, 5)$.

10. Find the slope of the line through $(1, -3)$ and $(5, 3)$.

11. Find the slope of the line through $(-2, 4)$ and $(4, -2)$.

12. Find the slope of the line through $(-5, -2)$ and $(3, 4)$.

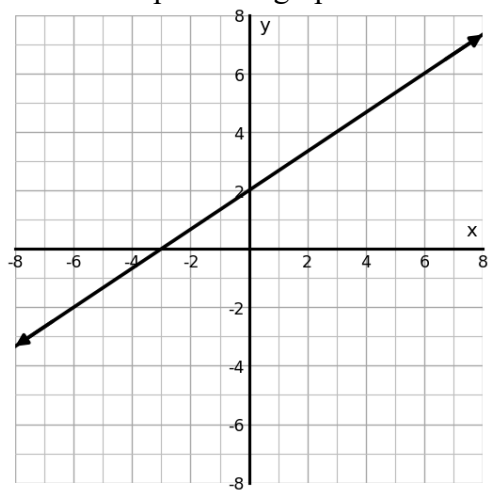
13. Are the given points collinear? Show how you determined your answer: $(-3, -2)$, $(1, 4)$, $(5, 10)$.

14. Are the given points collinear? Show how you determined your answer: $(-4, 5)$, $(0, 1)$, $(3, -1)$.

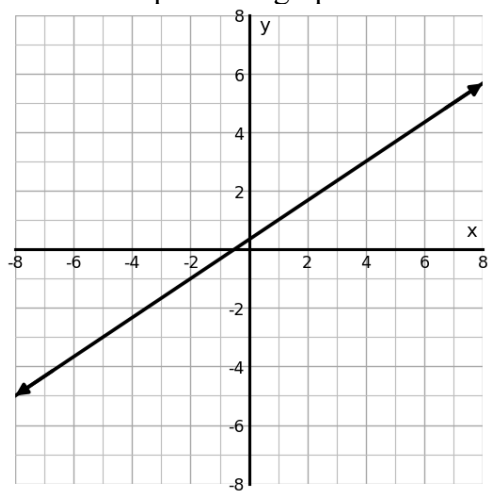
15. Are the given points collinear? Show how you determined your answer: $(-6, 7)$, $(-2, 1)$, $(4, -8)$.

16. Are the given points collinear? Show how you determined your answer: $(-5, -4)$, $(1, 2)$, $(4, 6)$.

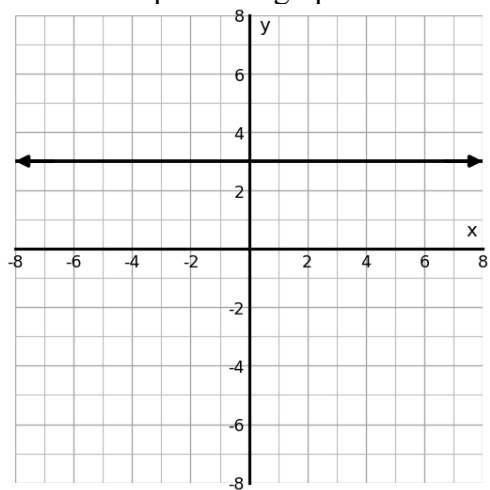
17. Find the slope of the graphed line.



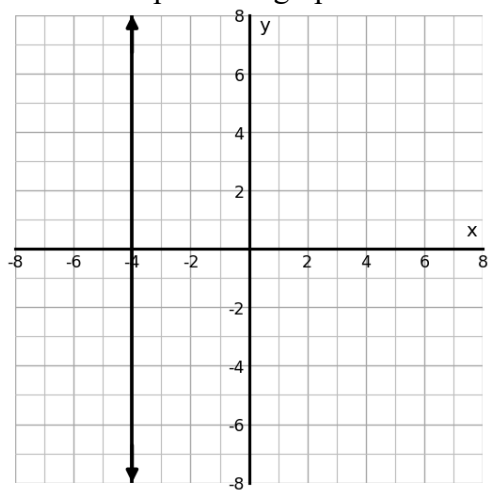
18. Find the slope of the graphed line.



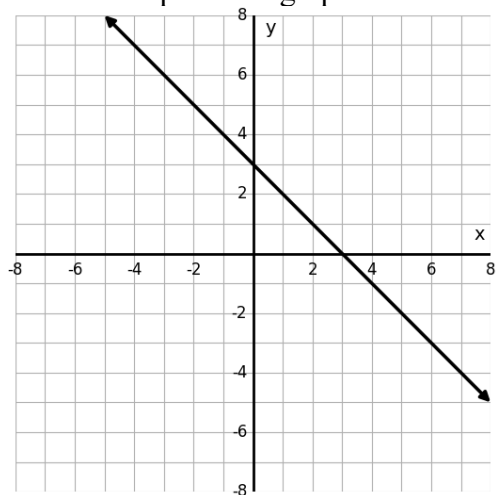
19. Find the slope of the graphed line.



20. Find the slope of the graphed line.



21. Find the slope of the graphed line.



22. Find the slope of the graphed line.

